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Experimental study of micromechanical properties alterations of shale matrix treated by $ScCO_2$-Water saturation using nanoindentation tests

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ABSTRACT

Supercritical CO_2 ($ScCO_2$) fracturing is a preferred alternative to water-based fracturing for stimulation of the shale gas reservoirs, and also promotes CO_2 geological sequestration. The injected $ScCO_2$ will interact with shale minerals and then induce changes in rock matrix micromechanics. In this study, a series of nanoindentation tests was conducted on the shale sample to investigate the effects of $ScCO_2$-water treatment on shale matrix micromechanics. X-ray diffraction (XRD), scanning electron microscopy (SEM) and energy-dispersive spectroscopy (EDS) were used to analyze the changing mechanisms of micromechanical properties. The dispersion of grid nanoindentation data of different minerals indicates the strong heterogeneity of shale' micromechanical properties. After $ScCO_2$-water treatment, different shale minerals experienced different reductions of E_r and H because of the dissolution induced the alterations in crystal structures, and their contents also altered in varying degrees. The values of h_f/h_m and W_p/W_t can be combined with the hysteresis loops enclosed by the P-h curves to determine the dominant work and irreversible deformation that produced in the process of indentation. According to the observation of indentation morphology, unlike other minerals, considerable micro-fractures were generated in clay minerals, which is correlated to the significant plastic deformation and the layered crystal structure.

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1. Introduction

Horizontal drilling and hydraulic fracturing have greatly promoted the development of shale gas. Water-based fracturing fluids are currently widely used to stimulate shale gas reservoir with ultra-low permeability by creating fracture networks due to easy availability and high economy [1]. However, various issues, including large water consumption, environmental risks and formation damage, limit the large-scale application of water-based fracturing, especially from clay-abundant shales and water-scarce areas [2–4]. Therefore, $ScCO_2$ (critical temperature is 31.1 °C and critical pressure is 7.38 MPa), a non-aqueous fracturing fluid, is considered to be a promising alternative because of its unique properties, including its ultralow viscosity, strong adsorption capacity and high density [5,6]. Compared with water fracturing, $ScCO_2$ fracturing not only avoids water consumption, but also reduces formation damage and induces more complicated fractures [7–9]. Moreover, it replaces adsorbed CH_4 for enhancing shale gas recovery and simultaneously facilitates CO_2 storage in shale formations due to its affinity for shale [10–12].

After $ScCO_2$ is injected into shale reservoirs, the CO_2-shale interaction will induce the alterations of mineral components and pore structures, and the mechanical properties as well as the flow behavior of both matrix and fractures could be altered [13–15], which ultimately influences the shale gas recovery. Therefore, it is essential to understand the effect of CO_2-shale interaction on the mechanical properties of shale. Researchers have conducted comprehensive uniaxial and triaxial compressive investigations of the evolution of the mechanical properties of shales after $ScCO_2$ exposure [16–19]. For example, Lyu et al. [16] found that, after saturation with $ScCO_2$-water for 30 days, the uniaxial compressive strength and Young's modulus of black shale decreased by 66.05% and 56.32%, respectively. Besides, Lyu et al. [20] also carried out UCS

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tests to investigate the stress-strain characteristics of shales with CO_2/CO_2-water treatment for 10 days, after treatment with $ScCO_2$, tests value slightly decreased by 4.62%, but the value for $ScCO_2$-water treatment declined by 38.87%. Similar results were also observed by Zhou et al. [19]. It can be seen that the effect of $ScCO_2$-water-shale interaction on the mechanical properties of shale is much more significant than that of $ScCO_2$-shale interaction. According to the statement of Gibbs [21]; the surface potential energy of shale would be reduced by the pure adsorbed CO_2, which causes the generation of shale matrix swelling and then leading to the reduction of mechanical strength. Hence, the mechanical strength reduction of shale in the imbibition condition of pure CO_2 is mainly because of the swelling caused by CO_2 adsorption, but in the imbibition condition of CO_2-water, the reduction is correlated to the combined reaction of swelling and dissolution. Consequently, it indicates that the influence of adsorption is significantly weaker on the mechanical strength of shales when compared with the effect of dissolution, so the focus of this study is the impact of dissolution on the mechanical properties of shale in the condition of $ScCO_2$ and water. However, to date most previous studies have been performed on inch-sized cores and have focused on their macro-mechanical variations, while study of micromechanical properties evolution related to different mineral compositions after $ScCO_2$-water saturation is scarce.

The nanoindentation technique has been widely used to characterize micromechanical properties because it requires small sample volume and short testing time and is a non-destructive instrumented method [22–27]. It was initially developed to study the micromechanical properties of homogeneous materials like metal or single crystal [28], but recently, nanoindentation testing has been introduced for various heterogeneous geomaterials, including shale, sandstone, coal, limestone and marble [24,29–31]. The measurement of micromechanical properties of formation rock is a significant work to predict the reservoir stimulation efficiency, such as the fracture initiation and propagation, fracture mechanical damage, proppant embedment behavior and so on. For instance, Yang et al. [32] conducted mineral-positioned nanoindentation on shale matrix minerals, results showed that quartz and feldspar exhibit significant elastic responses during the indentation process, while the combination of elastic and plastic deformation was observed for carbonates and clay minerals. They also found that the reliable mechanical properties of different shale matrix minerals, such as the Young's modulus and hardness of quartz, are 105–110 GPa and 14–16 GPa, respectively. Graham et al. [33] investigated the nano- and micro-mechanical properties of calcareous shale by combining atomic force microscopy and nanoindentation techniques, and found that the Young's moduli of quartz-feldspar, calcite and clay were 45–65 GPa, 34–50 GPa and 20–24 GPa, respectively. Shi et al. [34] utilized nanoindentation to investigate the effect of $ScCO_2$-water treatment on micro-mechanical properties of shale after exposure to a temperature of 70 °C and a pressure of 45 MPa for 48 h. The results indicated that the mechanical characteristics of treated samples were significantly weaker than those of untreated samples, and the effect of $ScCO_2$-water exposure in quartz-rich area was very small, whereas the reduction of mechanical properties in clay-rich area was significant. However, the geometry of the indenter applied in the study of Shi et al. [34] was relatively large and the applied load was also relatively high, causing different shale matrix minerals to be included in the indentation region, so it was difficult to investigate the mechanical properties of specific minerals. In addition, the extremely short exposure time was incapable of simulating the very long duration of field cases, leading to limited understanding of long-term caprock stability for CO_2 geological sequestration. Research on alterations to the micromechanical properties of specific shale minerals before and after $ScCO_2$-water exposure under high temperature and high pressure is also limited.

Therefore, in the study, we mainly investigate the alterations of micromechanical characteristics of shale matrix minerals before and after $ScCO_2$-water treatment under high-temperature and high-pressure conditions using the nanoindentation technique. Moreover, X-ray diffraction (XRD), scanning electron microscopy (SEM) and energy-dispersive spectroscopy (EDS) mapping were utilized to observe and identify the mineral changes and indentation morphology in different mineral phases due to shale-$ScCO_2$-water interaction. The alterations of the mechanical properties of shale matrix minerals can provide insights into the compressibility of fracture networks and the evolutionary laws of shale's micro-mechanical properties, and then revealing the inner weakening mechanisms of shale's macro-mechanical properties in the process of $ScCO_2$ fracturing and CO_2-enhanced shale gas recovery as well as the geological sequestration of CO_2.

2. Materials and methods

2.1. Sample preparation

The tested shale sample was collected from the marine Lower Silurian Longmaxi Formation in the Sichuan Basin, China. XRD results obtained using 200-mesh shale powder illustrated that the sample contained quartz (56.4%), calcite (20.9%), dolomite (16.0%), clay minerals (5.3%), and others (1.4%). For the nanoindentation tests, a shale sample was cored perpendicular to the bedding plane and then cut into a small piece about 8 × 8 × 3 mm (length × width × thickness). Since a smooth sample surface perpendicular to the indenter is important to obtain reliable results, both mechanical polishing and ion polishing were used, and the sample surface roughness should be less than one fifth of the average nanoindentation depth [35]. Both sample surfaces were ground and polished using an MP-1B lapping machine, perfectly parallel surfaces were realized by fixing the sample on a mold using paraffin, and a series of emery papers from 150 to 5000 mesh size was used. One shale sample surface was then selected to be finely polished using 0.5 and 0.05 μm diamond suspension polishing fluid successively until it was mirror-smooth. Ultrasonic cleaning in ethanol was used to remove small particles from the polished surface. Finally, a Leica EM RES102 (Fig. 1(b)) was utilized for polishing by argon ion milling at 4.0 kV intensity and a 3.0° angle for 5 h to ensure a very flat surface and control the undulation only in several nanoscale. The sample was dried at 60 °C for 48 h to remove the free moisture and then stored in a very elastic capsule to protect the sample and prevent contamination before the nanoindentation tests (Fig. 1(a)).

For the contrast experiments treated with $ScCO_2$-water, the same shale sample was saturated in a high-temperature and high-pressure reactor (YZRJ-1000, volume of 1L) (Fig. 1(c)). 150 mL deionized water and injected CO_2 were placed in the reaction chamber to simulate a weak acid environment like that in an actual shale gas reservoir, and after treatment at a temperature of 200 °C and pressure of 13.8 MPa for three months, the sample surface was repolished using diamond suspension polishing and argon ion polishing to remove the conductive coating and eliminate the effect of $ScCO_2$-water treatment on surface flatness.

2.2. Nanoindentation technique

2.2.1. Analytical theory of nanoindentation testing

The typical indentation test involves a linearly-increasing loading procedure, followed by a short holding stage and continues with a linearly-decreasing unloading procedure (Fig. 2(a)).

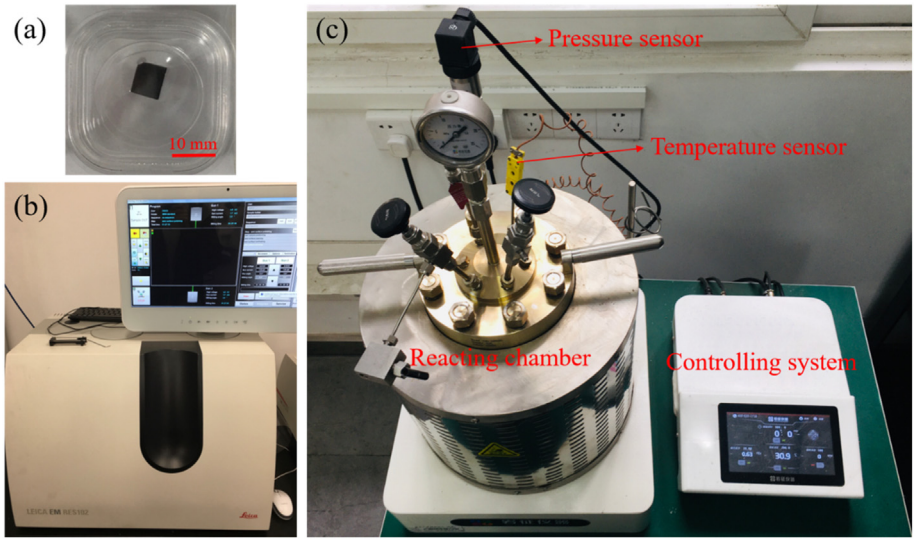


Fig. 1. Photographs of sample and apparatus: (a) small shale sample after polishing; (b) ion polishing equipment; (c) high-temperature and high-pressure reactor.

The load-displacement (P-h) curve can be obtained during the indentation process, and the mechanical parameters of the sample can be inferred according to the P-h response curve (Fig. 2(b)). The loading stage combines early elastic deformation and late plastic deformation, whereas the unloading stage is pure elastic deformation recovery.

The constant loading rate nanoindentation method is regarded as quasi-static nano-indentation, the methodology and operating principles of which have been reviewed in detail elsewhere [22]. According to the P-h curve (Fig. 2(b)), the mechanical parameters of indented materials can be derived using the Oliver and Pharr (O—P) method which is the classic data-interpreting method and has been widely applied in nanoindentation:

$$H = \frac{P_{\max}}{A_c} \tag{1}$$

where, H is the hardness, $P_{\max}$ is the maximum force and A_c is the contact area.

$$E_r = \frac{\sqrt{\pi}}{2\alpha}\frac{S}{\sqrt{A_c}} \tag{2}$$

where, E_r is the reduced Young's modulus, $S = \frac{dP}{dh}$ is the contact stiffness and is equal to the initial slope of unloading curve, and α is a constant related to the geometry of the indenter (1.034 for a Berkovich indenter).

The Young's modulus of the sample can be calculated by:

$$\frac{1}{E_r} = \frac{1-\nu^2}{E} + \frac{1-\nu_i^2}{E_i} \tag{3}$$

where, E and ν are the Young's modulus and Poisson's ratio of the indented sample, respectively, and E_i and ν_i are the Young's modulus and Poisson's ratio of the indenter, respectively. For the Berkovich diamond indenter, $E_i = 1140$ GPa and $\nu_i = 0.07$.

Furthermore, based on the P-h curve and the energy method, the total work W_t is the projection area $OABh_m$, and the elastic work

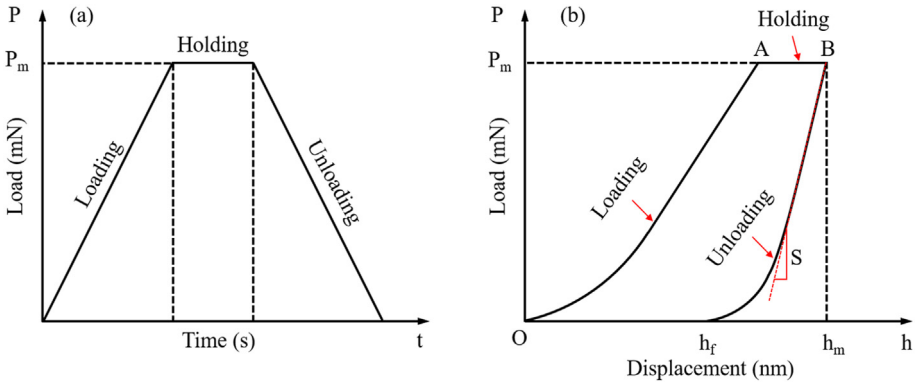


Fig. 2. Nanoindentation schematic diagrams: (a) variation of applied load with time; (b) typical load with displacement curve. h_m: maximum indentation depth. h_f: final indentation depth. S: the contact stiffness is equal to the slope of the initial unloading curve.

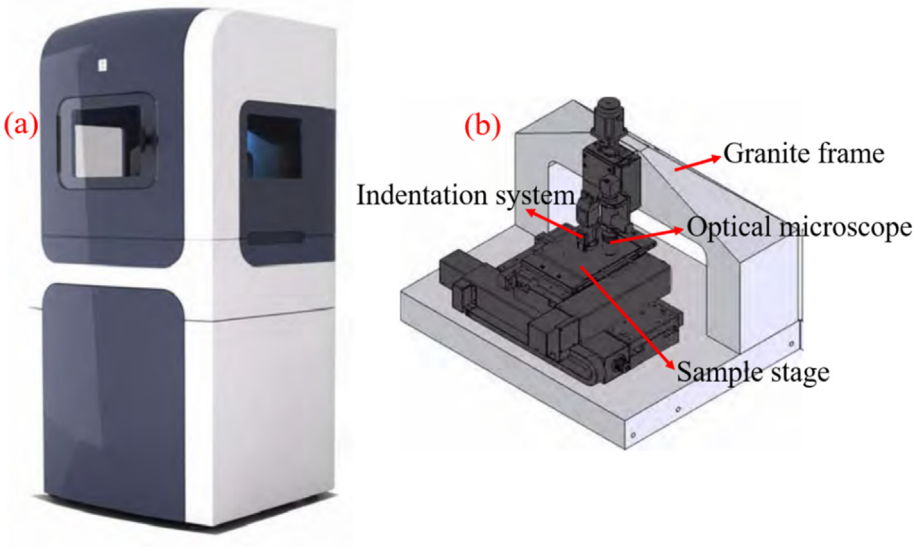


Fig. 3. Nano-indentation apparatus: (a) TI 950 Tribo-indenter; (b) major components of the apparatus.

W_e is the area underneath the unloading curve ($h_f Bh_m$) recovered by unloading. Therefore, the difference between the total work and the elastic work is defined as the plastic work W_p produced with loading, which can be calculated as:

$$W_p = W_t - W_e \tag{4}$$

2.2.2. Nanoindentation tests

In the present study, a Hysitron TI 950 Triboindenter was utilized for nano-indentation testing (Fig. 3(a)). This apparatus consists of four main parts: the granite frame supplying a ridged skeleton for components to be built on, an indentation system equipped with a Berkovich tip (a three-sided pyramidal diamond probe with a radius of curvature of 150 nm and a total included angle of 142.35°), an optical camera system with a standard 20× objective lens, and a default sample stage using magnetic attraction for securing samples (Fig. 3(b)). The sample stage is controlled by an automated X, Y, and Z-axis staging system, the travel distances of X and Y-axis are 300 mm × 300 mm, and the travel distance of Z-axis is 50 mm. The Triboscanner can be used to perform in-situ scanning probe microscopy (SPM) imaging of a sample to investigate qualitative and quantitative information about the testing area. A three-plate capacitive transducer developed by Hysitron is utilized to measure the displacement or force, and the data are acquired by a data acquisition computer. The maximum and minimum loads of the apparatus are 10 mN and 70 nN, respectively. The resolutions of the loads and displacements are 1 nN and 0.006 nm, respectively. In addition, the apparatus is equipped with a vibration isolation system and an acoustic enclosure to minimize disturbance from the ambient environment and noise. Before conducting nanoindentation tests, the testing accuracy was initially calibrated using a standard fused quartz.

For a heterogeneous material like shale, grid nanoindentation with a large number of indentation points is required to eliminate sampling bias and acquire statistical data, and the grid spacing should be larger than three times the imprint radius of the largest individual indentation to avoid interference between individual indentations and achieve statistical independence [23,36]. After the argon ion polishing process, indentation tests were conducted on the center of the shale sample with one 10 × 10 grid set and 5 μm spacing of each indentation point (Fig. 4). The sequence of each indentation point was defaulted, from the left to the right in the top row first (from point 0 to point 9), and the last indentation point was point 99 at the bottom right. The indenter was advanced at a constant rate of 1.6 mN/s to 8 mN and the loading time was 5 s, held at peak force for 2 s, and unloaded at 1.6 mN/s. After the indentation tests, in-situ SPM imaging was scanned with a frequency of 0.5 Hz and a low force of 2 μN. In order to further understand the effect of $ScCO_2$-water treatment on the micromechanical properties of different minerals, we studied the changes in the mineral components and the indentation surface morphology using a combination of XRD, SEM and EDS techniques. Each specific shale mineral has characteristic chemical elements, and with the help of the EDS technique, it can be exactly determined that the indentation region belongs to quartz, calcite, dolomite or clay minerals, and it can be

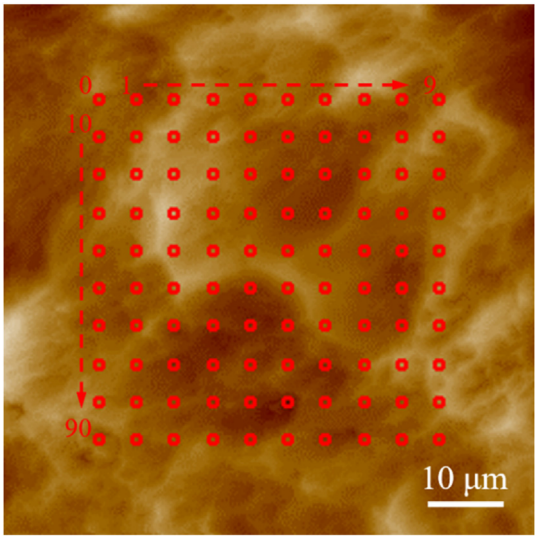


Fig. 4. Nano-indentation of 10 × 10 grid matrix with 5 μm grid spacing.

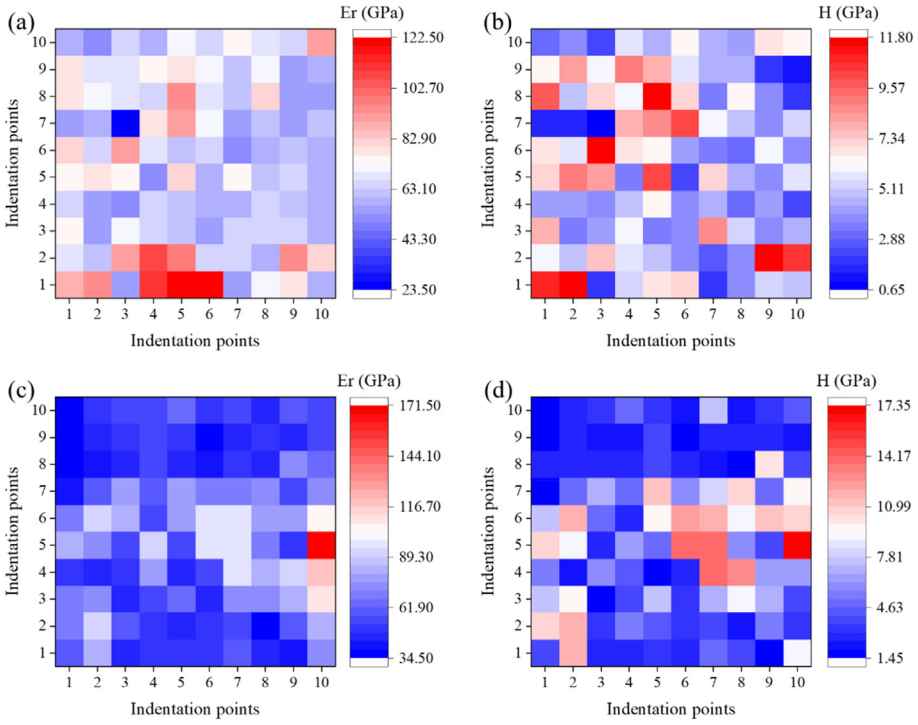


Fig. 5. Mechanical properties of grid indentation matrix: (a) reduced Young's modulus before $ScCO_2$-water treatment; (b) hardness before $ScCO_2$-water treatment; (c) reduced Young's modulus after $ScCO_2$-water treatment; (d) hardness after $ScCO_2$-water treatment.

double-checked by combining the mechanical parameters and the indentation surface morphology. Before conducting the SEM and EDS mapping tests, the sample surface was sprayed with a thin layer of gold for about 90 s to ensure the electrical conductivity of the shale sample and avoid charging effects.

3. Experimental results

The mechanical properties of the shale matrix before and after $ScCO_2$-water treatment were compared using nanoindentation testing on a 2025 μm^2 grid area. The E_r and H values of the shale sample for the grid nanoindentation test were calculated using Eqs. (1) and (2) and the results are shown in Fig. 5. As Fig. 5 shows, as the mechanical properties of individual indentations present significant discreteness, a sufficiently high number of nanoindentations is necessary to understand the heterogeneity of the mechanical properties. For the shale sample before $ScCO_2$-water treatment, the E_r have a wide range from 23.50 GPa to 122.50 GPa, and the H from 0.65 GPa to 11.80 GPa. The dispersion of the data is not only related to the flatness of the sample surface in the case when the surface roughness is excessively large [23], but to the heterogeneity of the shale sample. The maximum E_r and H of the treated sample were significantly larger than that of the untreated sample, mainly because of the nanoindentation which occurs in pyrite (FeS_2) with its stable crystal structure and extremely strong mechanical properties. Ignoring this exceptional indentation point, the average values of E_r and H after treatment were 62.69 GPa and 5.58 GPa, respectively, whereas the average values of E_r and H for the untreated shale sample were 68.73 GPa and 5.50 GPa, respectively. After treatment with $ScCO_2$-water, the average E_r of shale decreased by 8.79%, and the average H changed slightly. However, numerous mechanical behaviors, such as the initiation and propagation of induced fractures, are strongly associated with the mechanical properties of the weakest mineral phase in different shale mineral phases, but are not completely determined by the average mechanical properties of the overall shale. Therefore, the measurement of the mechanical properties of each specific shale mineral is also crucial.

Fig. 6 shows the P-h curves of the four main mineral components (quartz, calcite, dolomite and clay minerals) of the shale sample with and without $ScCO_2$-water treatment. The figure shows that while the maximum indentation depths (h_m) of different minerals vary under the same nanoindentation conditions, the h_m of clay minerals is greatest, its value being 400.93 nm, 2.02 times that of quartz before treatment. The h_m of quartz is lowest, being lower than the 308.49 nm of calcite and the 219.96 nm of dolomite before treatment, which is closely related to the different mechanical properties of different shale matrix minerals. The ranks of the final indentation depths (h_f) of different minerals are similar to those of the maximum indentation depths, regardless of the treatment condition. The P-h curves of calcite and clay minerals before treatment show two abnormal phenomena, namely "pop-in" and "elbow" patterns. A "pop-in" event, i.e., an abrupt increase of displacement when the load remains substantially constant during the loading process, is mainly caused when the indented tip encounters an original pore or micro-fracture in shale, or the cracks generated in the process of indentation [37]. The "elbow" phenomenon occurs in the unloading stage and shows a sudden decline of the slope of the unloading curve, the reason being the material expansion in the process of the slow transformation of different phases under the indented tip [38]. Fig. 6 also displays a P-h comparison of the shale before and after $ScCO_2$-water exposure in quartz, calcite, dolomite and clay minerals. The P-h curves of quartz almost coincide before and after $ScCO_2$-water treatment, and the calculated E_r and H change from 93.63 GPa to 11.79 GPa—93.66 GPa and 11.52 GPa for quartz, respectively, indicating that the $ScCO_2$-

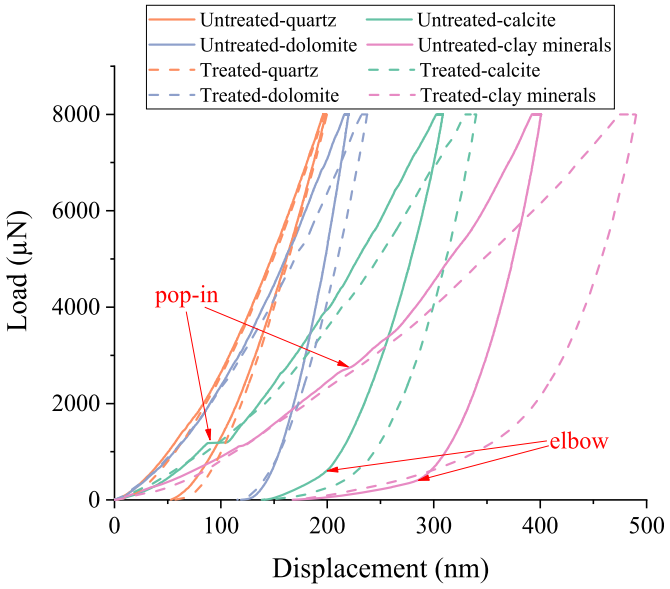


Fig. 6. Load-displacement curves of different minerals before and after $ScCO_2$-water treatment.

water treatment has less effect on the mechanical properties of quartz. Unlike quartz, the post-treated shale sample clearly shows an increased maximum indentation depth in carbonate and clay minerals. The greater nanoindentation depth indicates that a larger contact area, E_r and H values were calculated for the carbonate and clay minerals with $ScCO_2$-water treatment. Therefore, the mechanical properties of carbonate and clays can be expected to decrease after the interaction of $ScCO_2$, water and shale. The E_r and H of calcite decreased from 66.06 GPa to 3.83 GPa−62.64 GPa and 3.06 GPa, respectively. The E_r and H decreased from 121.98 GPa to 6.95 GPa−92.15 GPa and 6.56 GPa for dolomite. The E_r and H of clay minerals change from 56.35 GPa to 2.13 GPa−34.81 GPa and 1.50 GPa, respectively. According to the results, no apparent changes of mechanical properties for quartz occur before or after $ScCO_2$-water treatment. The E_r declines are in the proportion of 5.18%, 24.45% and 38.23% for calcite, dolomite and clay minerals, respectively. H shows a 20.10%, 5.61% and 29.58% reduction for calcite, dolomite and clay minerals, respectively.

4. Discussion

According to Figs. 5 and 6, different load-displacement curves reflect the strong heterogeneity of the micromechanical properties of shale minerals. The crystal structures of different minerals determine their mechanical responses. The lowest maximum indentation depth of quartz indicates that the Si−O bond forms a stable crystal structure, inducing strong mechanical strength and the chemical stability of quartz. For carbonate minerals, the different cation species in calcite (Ca^{2+}) and dolomite (Ca^{2+} and Mg^{2+}) lead to differences in bond energies with the same anion (CO_3^{2-}), and further induce changes in their crystal structures, which may be the dominant reason for the different mechanical properties of calcite and dolomite. The greatest indentation depth under the same indentation condition indicates the weakest mechanical strength of clay minerals in shale matrix minerals, which is mainly caused by the layered crystal structure, and the mechanical properties are dominated by the mineral contact between individual layered clay particles [23]. The degradation of the mechanical properties of shale matrix minerals due to the interaction of $ScCO_2$-water with shale was verified by conducting nanoindentation experiments. Although all shale minerals suffer from the $ScCO_2$-water environment, the effects of $ScCO_2$-water on the load-displacement curves for different minerals differ. Compared with quartz, the greater change of the maximum indentation depth of carbonate reflects the greater decrease of mechanical characteristics after $ScCO_2$-water treatment. The main reason for the weaker mechanical strength of carbonate following $ScCO_2$-water exposure can be explained by the effect of carbonate mineral dissolution. In particular, the effects on clay minerals are much greater than that on carbonate, mainly due to the combination of clay adsorption and the influence of the dissolution of $ScCO_2$. As previously mentioned, this study mainly focuses on the mechanical properties alterations of low-clay shale minerals caused by the impact of dissolution in the condition of $ScCO_2$ and water.

The decrease of the mechanical properties of the shale matrix in the $ScCO_2$-water environment is crucial for the compressibility of complex fracture networks and for the pore structures, and further influences shale gas recovery. Therefore, the evolution mechanism of the micromechanical properties of shale minerals via $ScCO_2$-water treatment is strongly related to the mineral composition and indentation fracture morphology.

4.1. Effect of mineralogy on mechanical properties

First, we conducted XRD analysis to determine the mineral composition of indented shale sample using Empyrean manufactured by PANalytical. From the same shale block, shale powders below 200 mesh-sized were subjected to XRD analysis to measure the alterations in the mineral components of shale due to the interaction of $ScCO_2$, water and shale. The Rietveld refinement [39]

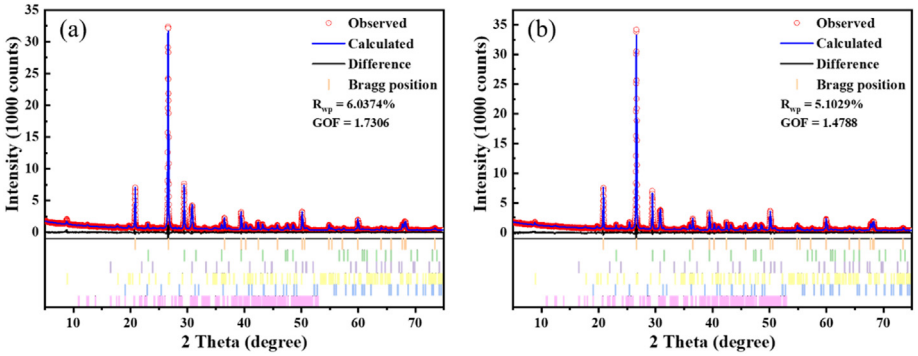


Fig. 7. Rietveld refinement results from the XRD patterns: (a) before treatment; (b) after treatment.

Table 1
Mineral compositions of different shale samples from the Rietveld refinement (%).

Conditions	Quartz	Calcite	Dolomite	Clay minerals	Others
Untreated	56.4	20.9	16.0	5.3	1.4
$ScCO_2$-water	57.7	18.8	15.5	4.5	3.5

Note: Clay minerals are the total amount of montmorillonite, illite, kaolinite and chlorite; other minerals include pyrite, anhydrite, etc.

was applied in this study to quantitatively analyze the proportion of different phases. As shown in Fig. 7, the diffraction patterns of two samples are in good agreement with the calculated ones, and the Bragg positions represent the positions of diffraction peaks of one specific mineral in shale. The refinement process was verified by two indexes, R_{wp} (weighted R profile) and GOF (goodness of fit), both of which were shown in the corresponding diffractogram. The mineral components of untreated and $ScCO_2$-water treated shale samples could be obtained from the refinement results as shown in Table 1. According to the XRD results, the shale sample is mainly composed of quartz (56.4%), calcite (20.9%), dolomite (16.0%) and clay minerals (5.3%), and the crystal systems for quartz, calcite and dolomite are all hexagonal, but the space groups are $P3_121$, R-3c and R-3, respectively. According to previous studies [40–42], following exposure to the $ScCO_2$-water environment, the formation of carbonic acid (see Eq. (5)) induces different complex chemical reactions with shale matrix minerals, and the chemical reaction equations between carbonic acid and different minerals are presented using Eqs. 6–11:

$$CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^- \quad (5)$$

$$quartz + 4H^+ \leftrightarrow Si^{4+} + H_2O \quad (6)$$

$$calcite + 2H^+ \leftrightarrow Ca^{2+} + CO_2 + H_2O \quad (7)$$

$$dolomite + 4H^+ \leftrightarrow Ca^{2+} + Mg^{2+} + 2CO_2 + 2H_2O \quad (8)$$

$$illite + 8H^+ \leftrightarrow 0.6K^+ + 2.3Al^{3+} + 0.25Mg^+ + 3.5SiO_2 + 5H_2O \quad (9)$$

$$kaolinite + 6H^+ \leftrightarrow 2Al^{3+} + 2SiO_2 + 5H_2O \quad (10)$$

$$chlorite + 16H^+ \leftrightarrow 5Fe^{2+} + 2.3Al^{3+} + 3SiO_2 + 12H_2O \quad (11)$$

Table 1 shows that the content of quartz remains almost the same and even slightly increases after $ScCO_2$-water exposure due to its strong chemical stability; in contrast, the dissolution of carbonates (including calcite and dolomite) and clay minerals promotes a decrease in content. This is consistent with the findings of other studies [43,44]. After $ScCO_2$-water treatment, the carbonate and clay mineral contents decrease from 36.9% to 5.3%–34.3% and 4.5%, respectively. The dissolution of carbonate and clay minerals in carbonic acid indirectly increases the weight ratio of quartz. Moreover, based on reaction Eqs. 9–11, the clay minerals can be converted into quartz under CO_2-water conditions, which may also account for the increase of the quartz content in shale. The increase of other mineral compositions indicates the recrystallization of dissolved minerals.

Table 2
Indentation data for different minerals before and after treatment.

Minerals	Conditions	h_f (nm)	h_m (nm)	h_f/h_m	W_p (pJ)	W_t (pJ)	W_p/W_t
Quartz	Untreated	58.45	198.70	0.29	174.43	622.15	0.28
	$ScCO_2$-water	70.47	199.66	0.35	181.03	607.70	0.30
Calcite	Untreated	155.03	308.49	0.50	555.63	952.40	0.58
	$ScCO_2$-water	180.01	340.05	0.53	708.57	1101.62	0.64
Dolomite	Untreated	126.96	219.96	0.58	430.65	703.38	0.61
	$ScCO_2$-water	123.16	237.63	0.52	446.40	791.69	0.56
Clay minerals	Untreated	249.92	400.93	0.62	791.64	1166.42	0.68
	$ScCO_2$-water	238.28	490.13	0.49	1097.67	1655.07	0.66

According to Eq. (3), in order to acquire the calculated E of the four main mineral constituents before and after treatment, the Poisson's ratios of each specific shale mineral need to be known. In the present study, on the basis of previous studies [45,46], the classic values of Poisson's ratios for typical minerals in shale were adopted, and the Poisson's ratios for quartz, calcite, dolomite and clay minerals are 0.06, 0.28, 0.29 and 0.30, respectively. Therefore, the E of each specific shale mineral before and after treatment could be calculated using Eq. (3) and the results are presented in Fig. 8. The E of quartz remains basically unchanged, due to the minimal effect of $ScCO_2$-water on quartz, while the reductions in calcite, dolomite and clay minerals reach up to 5.48%, 26.59% and 39.42%, respectively. The dissolution of carbonate and clay minerals changes the crystal structure and induces the decrease of micromechanical properties, and especially for clay minerals with strong layered structures [23], the dissolution of mineral contacts increases the porosity and compressibility among different layers [47], and therefore induces the maximum reduction in micromechanical parameters. Another possible reason for clay minerals is the adsorption of $ScCO_2$ [34], which further changes the crystal structure, ultimately leading to the reduction of the mechanical properties of clay minerals. According to Fig. 8 (a) and (b), for two carbonate minerals in shale, the E reduction of dolomite is

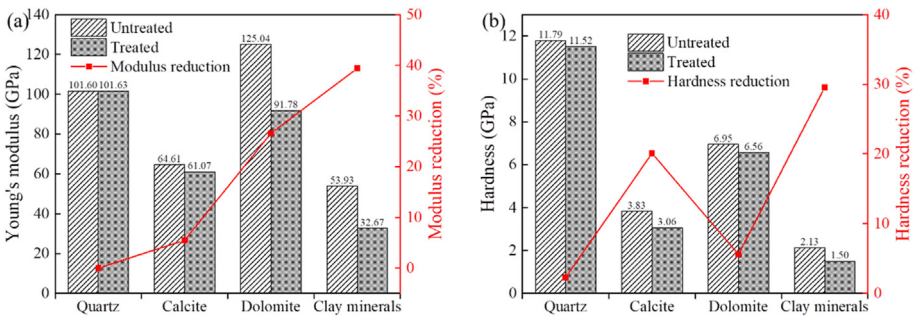


Fig. 8. Mechanical parameters of different minerals with and without $ScCO_2$-water treatment based on nanoindentation tests: (a) Young's modulus and its reduction; (2) hardness and its reduction.

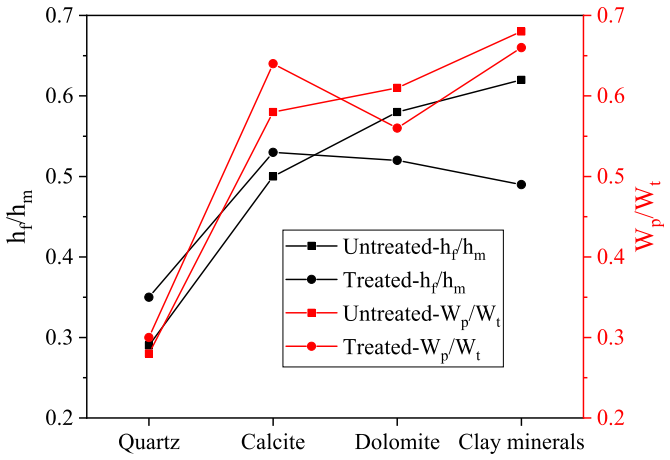


Fig. 9. Elastic recovery ratios and plastic work ratios for different minerals with and without treatment.

significantly greater than that of calcite, whereas the H reduction shows an opposite trend, which can be explained by the different crystal structures of these two carbonate minerals, as stated previously.

In order to further understand the alterations of energy during the nanoindentation tests, we compared the elastic recovery ratios (h_f/h_m) and plastic work ratios (W_p/W_t) before and after $ScCO_2$-water treatment based on Eq. (4) and P-h curves (Fig. 6), and the results are presented in Table 2 and Fig. 9. According to Fig. 9, the h_f/h_m and W_p/W_t of quartz are lowest, being smaller than 0.40 compared to other minerals under $ScCO_2$-water treated or untreated conditions, and mainly depend on the stable crystals and strong micromechanical properties of quartz. For quartz and calcite, the values of h_f/h_m and W_p/W_t before treatment are 0.29, 0.28, 0.50 and 0.58, respectively, which are smaller than the post-treatment values. However, the values of h_f/h_m and W_p/W_t for dolomite and clay minerals are reduced after treatment with $ScCO_2$-water. According to Table 1, there are minor amounts (5–25%) of dolomite and traces (<5%) of clay minerals in the shale sample after treatment, and the indented phase (dolomite or clay minerals) under the indenter may slowly transfer to a harder phase (quartz, calcite or dolomite) during the unloading stage in the nanoindentation tests, leading to the decrease of h_f in dolomite and clay minerals after treatment, ultimately inducing the decrease of h_f/h_m and W_p/W_t.

On the basis of the combination of Figs. 6 and 9, the alterations of energy during nanoindentation tests can be shown quantitatively. Before treatment, the small values of h_f/h_m and W_p/W_t in quartz illustrate that elastic work is dominant and irreversible deformation is small in the process of indentation, and the extremely narrow hysteresis loop surrounded by P-h curve also supports this. For carbonate minerals before exposure, the h_f/h_m and W_p/W_t range from 0.50 to 0.58 and 0.58–0.61, respectively, indicating that there is more plastic work and greater irreversible deformation during indentation. Correspondingly, the areas of the hysteresis loops enclosed by the P-h curves of carbonate minerals are larger than those for quartz. The calculated h_f/h_m and W_p/W_t for untreated clay minerals are 0.62 and 0.68, respectively, showing that the total work in the process of indentation is dominated by plastic work and there is significant irreversible deformation. The largest area of the hysteresis loop produced by the P-h curve further supports this. For calcite, both values of h_f/h_m and W_p/W_t increase significantly after $ScCO_2$-water treatment, indicating that plastic work and irreversible deformation are increased, caused mainly by the decline of micromechanical properties after treatment.

4.2. Indentation morphology

SPM images of untreated and $ScCO_2$-water treated shale samples obtained using Triboscanner after nanoindentation tests (Fig. 10) show that the indentation geometries are different in different regions. In general, the hard mineral phases (silicate and carbonate) can be polished easily and soft mineral phases (clay minerals and organic matter) tend to be pulled out, causing the surfaces of hard phases to be very flat while the soft phases are rougher [32]. At the same time, the indentation geometries of hard phases are smaller than those of soft phases. Therefore, smaller indentation depths and flat surfaces are responses to hard phases, and conversely, greater indentation depths and rough surfaces are responses to soft phases. The indentation morphology of a specific mineral was therefore further analyzed using SEM and EDS techniques.

The indentation morphologies and mineral compositions were analyzed using SEM/EDS techniques. The different minerals in shale can be identified by EDS mapping, since each mineral has a particular chemical or elemental composition. For example, quartz

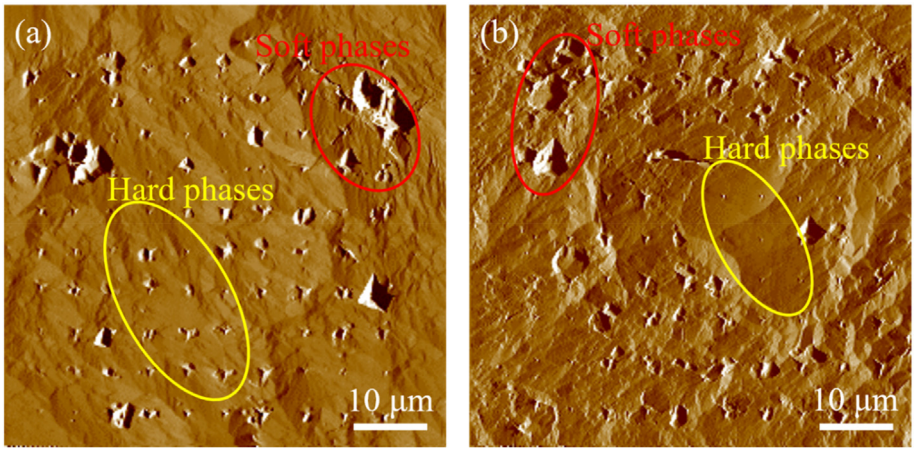


Fig. 10. SPM images after nanoindentation: (a) untreated sample; (b) $ScCO_2$-water treated sample.

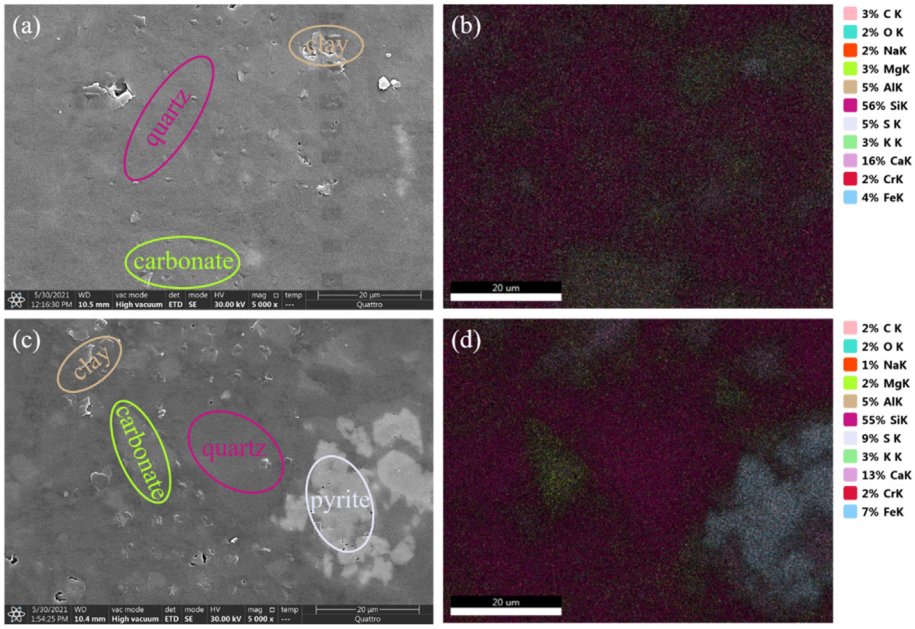


Fig. 11. SEM and EDS mapping photographs of grid indentation matrix: (a) and (c) SEM photographs before and after treatment; (b) and (d) EDS mapping photographs before and after treatment.

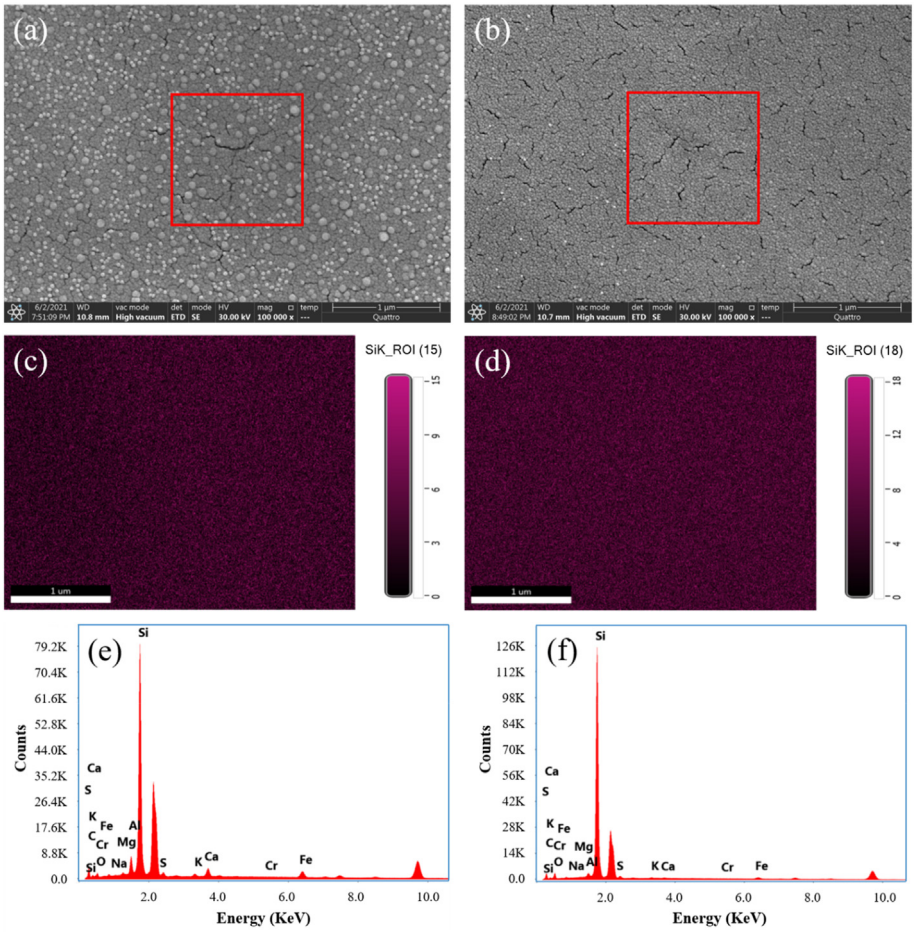


Fig. 12. SEM and EDS mapping photographs of post-indented quartz before and after ScCO2-water treatment: (a) and (b) SEM photographs before and after treatment; (c) and (d) Si element distribution before and after treatment; (e) and (f) EDS mapping results before and after treatment.

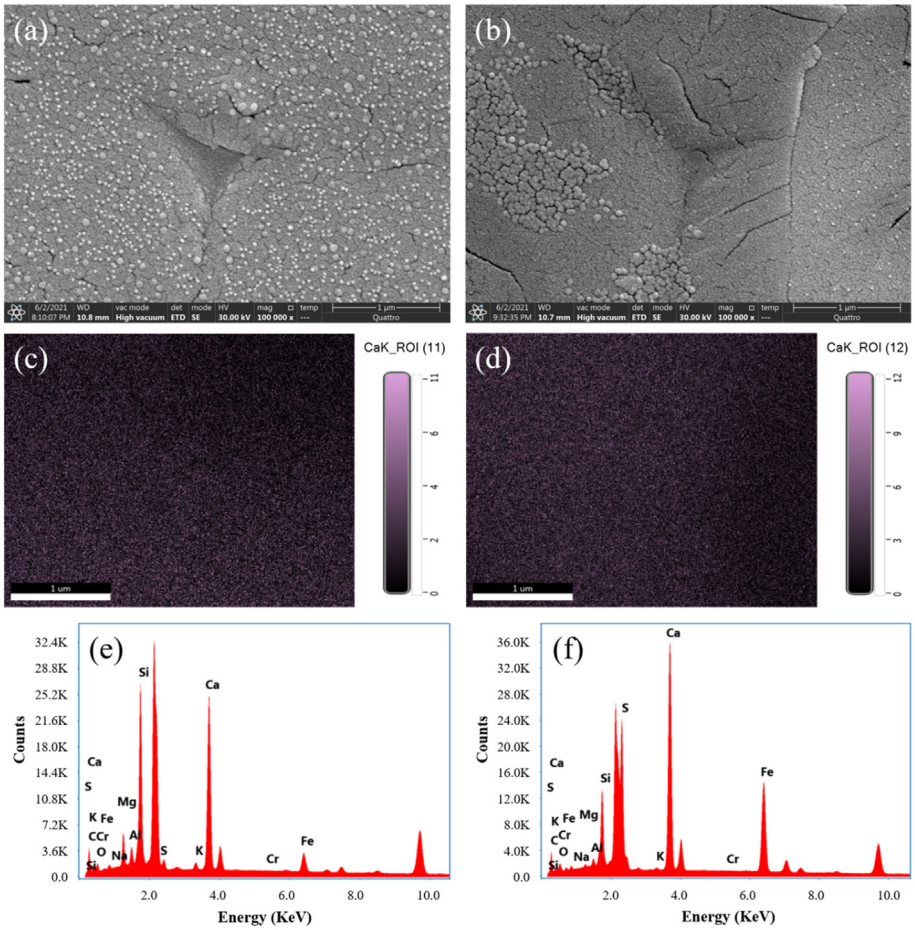


Fig. 13. SEM and EDS mapping photographs of post-indented calcite before and after ScCO2-water treatment: (a) and (b) SEM photographs before and after treatment; (c) and (d) Ca element distribution before and after treatment; (e) and (f) EDS mapping results before and after treatment.

is composed of silicon and oxygen atoms, the characteristic elements of carbonate are calcium and magnesium, while clay minerals are hydrous aluminium phyllosilicates [48]. We can compare the nanoindentation morphology differences in different minerals with the help of EDS, and the indentation depths and mechanical properties can be combined to confirm that the measurement area relates to quartz, carbonate or clay minerals. Fig. 11 shows SEM and EDS mapping images of a nanoindentation grid matrix before and after $ScCO_2$-water treatment. In Fig. 11 (b) and (d), the elemental distributions of shale with and without $ScCO_2$-water treatment are shown, and the abundance of Si reaches up to 56%, the minor element is Ca at about 13–16%, and there are traces of other elements, including Mg and Al. It can be inferred that the major mineral in the nanoindentation area is quartz by combining the results of XRD analysis, the presence of Ca and Mg elements can be used to find carbonate minerals, and clay minerals can be identified by the presence of Al (see Fig. 11 (a) and (b)). The indentation fracture geometry is greatly correlated with the mechanical properties of minerals, which can be used to confirm the measurement regions, and the results of grid nano-indentation tests (Fig. 5) in the corresponding regions also support the credibility of this inference.

Figs. 12–15 display SEM and EDS mapping images of single nanoindentations in quartz, calcite, dolomite, and clay minerals with and without $ScCO_2$-water treatment. In (a) and (b) of each of Figs. 12–15, the small particles on each sample surface are gold particles because each shale sample was sprayed with gold before conducting SEM and EDS analysis, and the original micro-fractures of shale can be easily found because of the extremely high magnification applied in the SEM tests. The distributions of the characteristic elements of each shale matrix mineral are shown in (c) and (d) in each of Figs. 12–15, and can be used to identify the specific minerals. In addition, (e) and (f) in each of Figs. 12–15 show the entire elements and their intensities in relation to the corresponding SEM images in (a) and (b). It can be easily observed that each nanoindentation morphology on different minerals under the same indentation conditions has different indentation types due to the differences of microstructure and mechanical properties. Fig. 12 shows that there are no distinct indentation fractures on quartz, with or without $ScCO_2$-water treatment. This can be explained by the fact that quartz generates elastic work and recoverable deformation due to its strong mechanical properties and chemical stability, and the small loading pressure applied during the nanoindentation tests was incapable of producing greater irreversible deformation. However, according to Figs. 13 and 14, the indentation fractures in carbonate (calcite and dolomite) are more obvious than those in quartz, and the indentation fracture geometries in calcite are larger than those in dolomite under the same conditions. This can be explained by the greater mechanical properties of dolomite than calcite due to the different crystal structures of the two carbonate minerals. In addition, the fracture propagation imprint of carbonate with $ScCO_2$-water treatment is much more obvious than that of carbonate without treatment, demonstrating

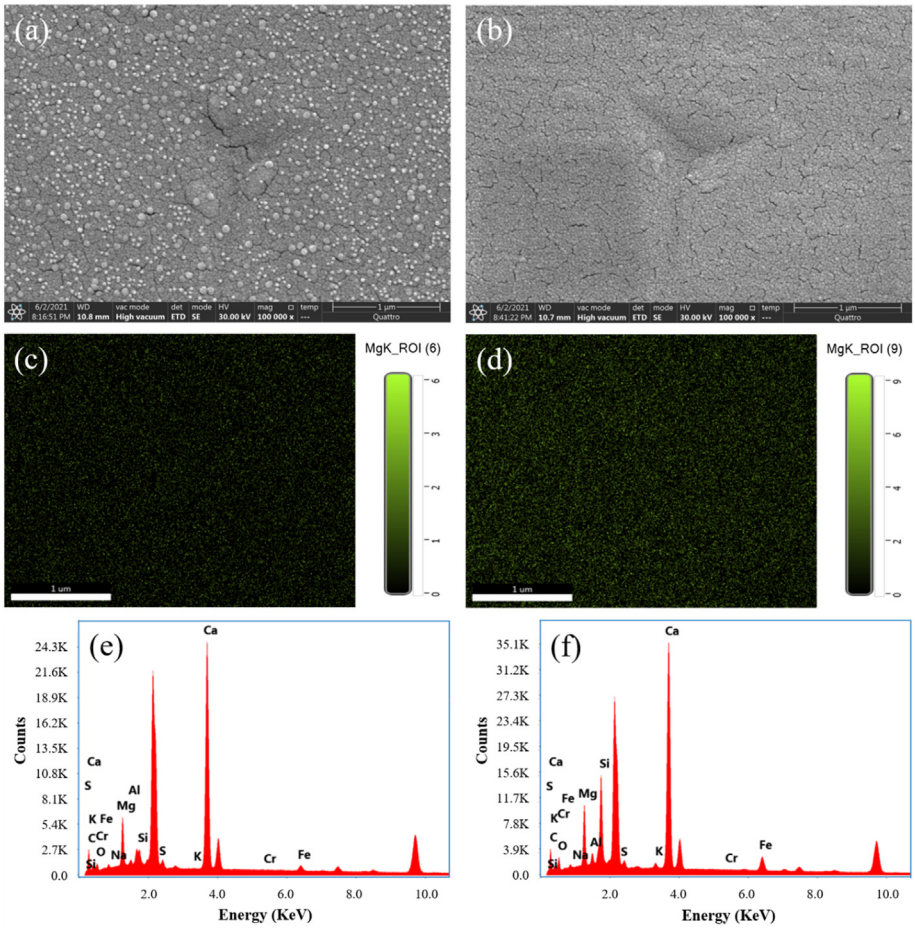


Fig. 14. SEM and EDS mapping photographs of post-indented dolomite before and after ScCO2-water treatment: (a) and (b) SEM photographs before and after treatment; (c) and (d) Mg element distribution before and after treatment; (e) and (f) EDS mapping results before and after treatment.

that more plastic work and greater irreversible deformation were produced in the process of indentation, mainly because of the decrease of mechanical properties induced by the partial dissolution of minerals. For clay minerals, Fig. 15 indicates that many micro-fractures occur under untreated and post-treated conditions. The reasons for this phenomenon can be divided into two parts: the greatest irreversible deformation and significant plastic work were generated during the nano-indentation tests due to the weakest mechanical strength in shale matrix minerals; clay minerals are composed of phyllosilicates [48] and the mechanical properties are dominated by mineral contact between individual layered clay particles [23]. The relative displacement occurred between individual clay layers when force was applied, inducing the layers to bend and slip, but without generating internal stress. When the applied force was removed, the bent and slipped clay layers were incapable of returning to their original position due to the lack of internal stress, finally producing considerable plastic deformation. Furthermore, a greater number of micro-fractures were generated by $ScCO_2$-water treatment, the main reason being mineral contact dissolution leading to increased porosity and compressibility among different clay layers [47], ultimately inducing the greater reduction of the mechanical properties of clay minerals. Another possible reason for this finding may be the crystal structure change due to $ScCO_2$ adsorption in clay minerals [34]. Importantly, the fracture propagation path varies in a local region because of the heterogeneity of the shale matrix.

5. Conclusions

In this study, the evolution of the micromechanical properties of shale matrix minerals following $ScCO_2$-water treatment was investigated using nanoindentation tests. According to the experimental results, the micromechanical characteristics of specific shale minerals at nano- or micro-scale can be effectively determined by using the nanoindentation method, which is helpful for the estimation of the overall mechanical response after Sc-CO_2 treatment. The following conclusions can be drawn:

1) The micromechanical properties of four main shale mineral constituents can be tested by the nanoindentation technique. The values of E_r and H of quartz, calcite, dolomite and clay minerals were obtained from the P-h curves of each indentation, before $ScCO_2$-water treatment, the E_r of them are 93.63 GPa, 66.06 GPa, 121.98 GPa and 56.35 GPa separately, and the H for them are 11.79 GPa, 3.83 GPa, 6.59 GPa and 2.13 GPa, respectively. The dispersion of grid nanoindentation data indicates the strong heterogeneity of shale's micromechanical properties.
2) The XRD results show that the alterations occur in mineral components of shale in the weak acid environment of $ScCO_2$-water treatment due to different chemical reactions. The carbonate and clay mineral contents are decreased from 36.9% to 5.3%–34.3% and 4.5%, respectively, which indirectly increases the content of quartz from 56.4% to 57.7%, another potential

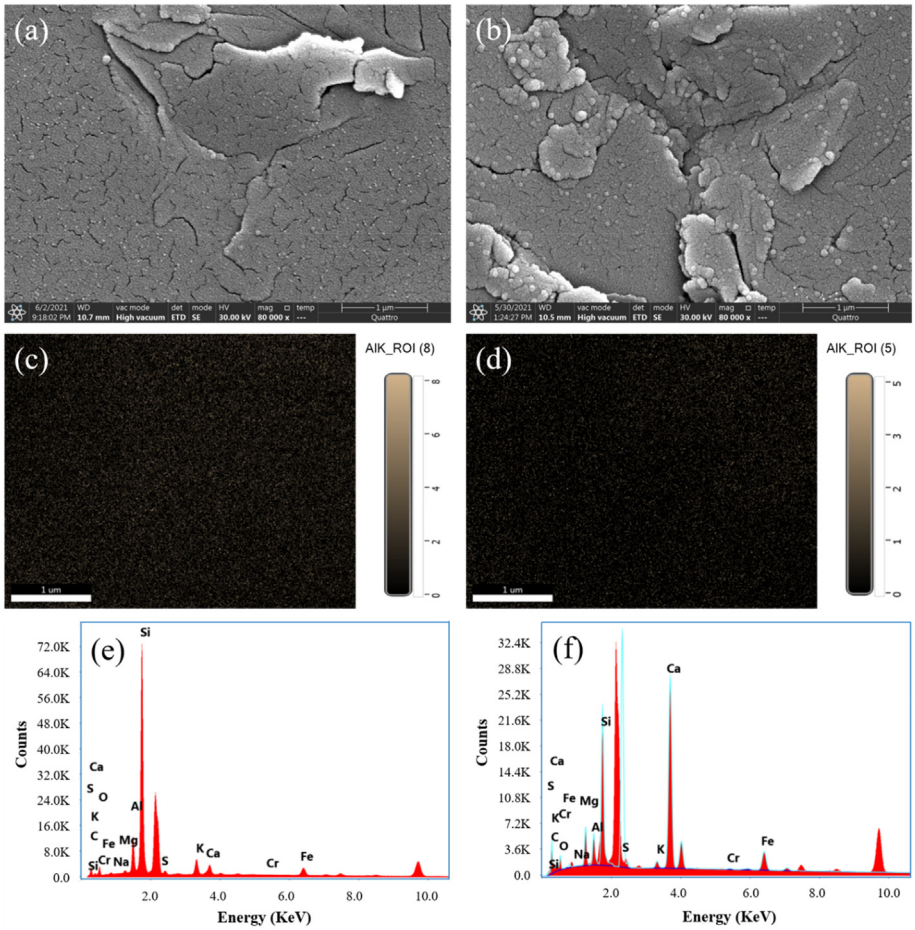


Fig. 15. SEM and EDS mapping photographs of post-indented clay minerals before and after ScCO2-water treatment: (a) and (b) SEM photographs before and after treatment; (c) and (d) Al element distribution before and after treatment; (e) and (f) EDS mapping results before and after treatment.

reason for the raised weight ratio of quartz is the conversion of clay minerals under the CO_2-water condition.

3) After treatment with $ScCO_2$-water, the average value of E_r in the grid indentation region is decreased. The values in specific shale minerals are also declined except for the quartz because of its strong mechanical strength and chemical stability. The E_r reductions are in the proportion of 5.18%, 24.45% and 38.23% for calcite, dolomite and clay minerals, respectively, and the weaker mechanical strength of shale after $ScCO_2$-water exposure is mainly correlated to the changes in micromechanical properties of dolomite and clay minerals.
4) The h_f/h_m and W_p/W_t can be used to combine with the hysteresis loops enclosed by the P-h curves to analyze the dominant work and irreversible deformation that produced in the process of indentation. The small values of h_f/h_m and W_p/W_t in quartz and the narrow hysteresis loop illustrate that the elastic work is dominant and the produced irreversible deformation is small, while for clay minerals, the situation is opposite.
5) According to the observation of indentation morphology, there are no obvious indentation fractures on quartz with or without $ScCO_2$-water treatment due to its strong mechanical properties and chemical stability. However, for carbonate, $ScCO_2$-water treatment causes larger indentation fracture geometries, because the dissolution of carbonate induces the decrease of mechanical properties. Much more abundant micro-fractures were generated and greatest indentation depth was occurred in clay minerals than in other minerals, and this can be explained as being a result of the combined effects of the largest irreversible deformation and the layered crystal structure.

Credit author statement

P Cheng: Investigation, Data curation, Writing – original draft. CP Zhang: Conceptualization, Methodology, Resources, Supervision, Funding acquisition. ZY Ma: Visualization, Methodology. JP Zhou: Data Curation, Writing – review & editing. DC Zhang: Writing - Review & Editing. XF Liu: Writing - Review & Editing. H Chen: Investigation. PG Ranjith: Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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